



QUBIT-BASED FUZZY RULE INFERENCE FOR DYNAMIC SYSTEM CONTROL : AN INVERTED PENDULUM

डायनामिक सिस्टम नियंत्रण के लिए क्यूबिट-आधारित फ़ज़ी नियम इन्फ़रेंस: एक उलटा पेंडुलम

UNDER GUIDANCE OF

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INTRODUCTION & MOTIVATION

THE CONTROL CHALLENGE:

- INVERTED PENDULUM: A CLASSIC, INHERENTLY UNSTABLE, NON-LINEAR SYSTEM.
- REQUIRES ROBUST STABILIZATION AGAINST DYNAMIC DISTURBANCES.

CONVENTIONAL CONTROL LIMITATIONS:

- PID/STATE-SPACE: DEMANDS PRECISE MODELS; STRUGGLES WITH UNCERTAINTY.
- BOOLEAN LOGIC: LACKS NUANCE FOR VAGUE, REAL-WORLD CONDITIONS.

FUZZY LOGIC ADVANTAGE:

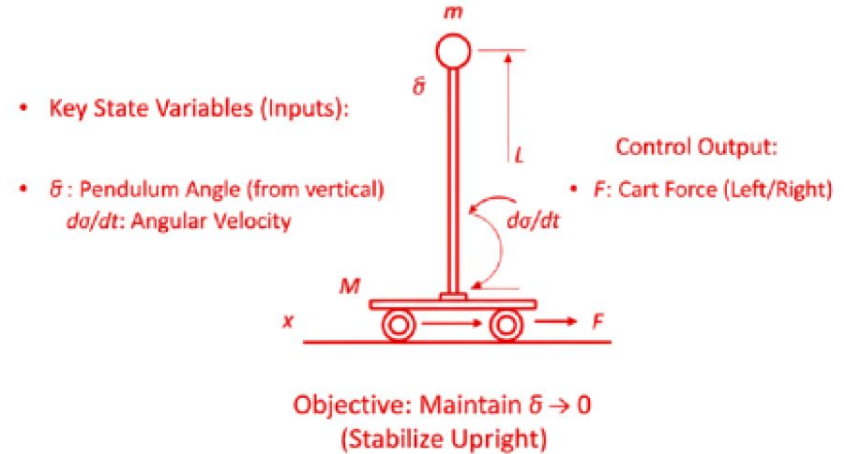
- MODELS HUMAN-LIKE REASONING AND INTUITION.
- ENABLES DECISION-MAKING WITH IMPRECISE DATA (DEGREES OF TRUTH).

QUANTUM COMPUTING POTENTIAL:

- LEVERAGES SUPERPOSITION/ENTANGLEMENT FOR NOVEL INFERENCE ARCHITECTURES.
- EXPLORES FUTURE COMPUTATIONAL ADVANTAGES IN COMPLEX CONTROL.

PROJECT OBJECTIVE:

- DEVELOP A QUANTUM FUZZY INFERENCE ENGINE (QFIE) FOR INVERTED PENDULUM STABILIZATION.
- DEMONSTRATE SYNERGISTIC INTEGRATION OF FUZZY LOGIC AND QUANTUM COMPUTING.



INVERTED PENDULUM SYSTEM

THE INVERTED PENDULUM SYSTEM

KEY STATE VARIABLES (INPUTS):

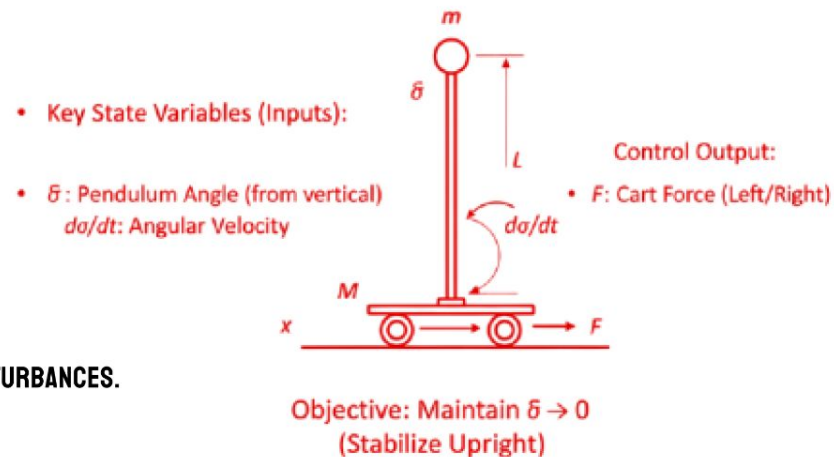
- PENDULUM ANGLE (Θ): DEVIATION FROM VERTICAL.
- ANGULAR VELOCITY ($D\Theta/DT$): RATE OF CHANGE OF ANGLE.

CONTROL OUTPUT:

- CART FORCE (F): FORCE APPLIED TO THE CART (LEFT/RIGHT).

THE CONTROL OBJECTIVE:

- TO KEEP THE ANGLE AS CLOSE TO ZERO (VERTICAL) AS POSSIBLE, REGARDLESS OF DISTURBANCES.



INVERTED PENDULUM SYSTEM

FUZZY LOGIC: HUMAN-LIKE REASONING IN CONTROL

DEFINITION:

- A MULTI-VALUED LOGIC DEALING WITH "DEGREES OF TRUTH" (0-1), NOT JUST STRICT TRUE/FALSE.
- MODELS HUMAN INTUITION FOR VAGUE CONCEPTS.

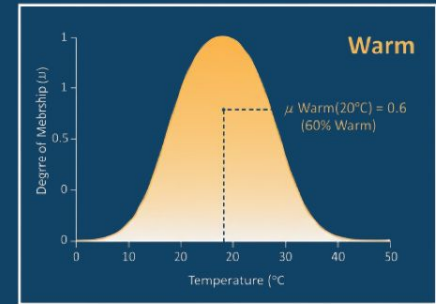
CORE PRINCIPLE: PARTIAL MEMBERSHIP:

- VALUES CAN BELONG TO A SET PARTIALLY (E.G., 60% "POSITIVE ANGLE," 40% "ZERO ANGLE").
- AVOIDS RIGID BOUNDARIES OF CLASSICAL LOGIC.

BENEFITS FOR CONTROL SYSTEMS:

- ROBUSTNESS: HANDLES UNCERTAINTY AND IMPRECISION EFFECTIVELY.
- SMOOTH CONTROL: ENABLES GRADUAL, NUANCED RESPONSES.
- INTUITIVE DESIGN: RULES ARE EXPRESSED IN NATURAL LANGUAGE.

Fuzzy Logic: Membership Function Example



A membership function defines the "degree of truth" for the fuzzy set.

FUZZIFICATION:

- CONVERTING EXACT MEASUREMENTS INTO DEGREES OF MEMBERSHIP FOR FUZZY SETS.

RULE INFERENCE:

- APPLYING FUZZY RULES (IF-THEN STATEMENTS) TO DERIVE FUZZY CONCLUSIONS.

DEFUZZIFICATION:

- CONVERTING THE FUZZY CONCLUSIONS BACK INTO A SINGLE, ACTIONABLE CRISP OUTPUT.

MEMBERSHIP FUNCTIONS: QUANTIFYING VAGUENESS

DEFINITION:

- MAP A CRISP INPUT VALUE (x) TO ITS DEGREE OF MEMBERSHIP (μ) $[0, 1]$ WITHIN A FUZZY SET.
- CORE MECHANISM FOR FUZZIFICATION.

IMPORTANCE:

- ENABLE PARTIAL BELONGING, NOT JUST BINARY (IN/OUT).
- FACILITATE SMOOTH TRANSITIONS IN CONTROL RESPONSES.
- SELECTED SHAPES: TRIANGULAR (TRIMF) & TRAPEZOIDAL (TRAPMF)

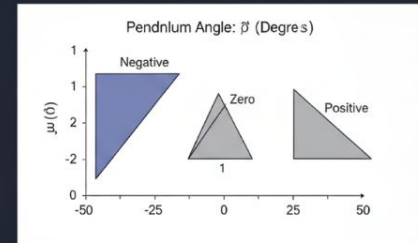
RATIONALE:

- SIMPLICITY, COMPUTATIONAL EFFICIENCY, AND INTUITIVE REPRESENTATION FOR CONTROL VARIABLES.

Membership Functions: Quantifying Vagueness

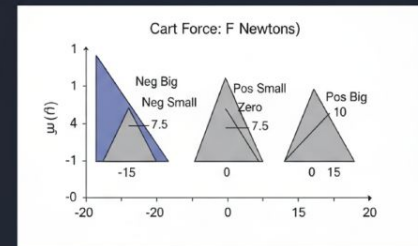
Definition:

- Membership functions map a crisp input to degree of membership between 0 1 fuzzification.
the core mechanism for



Importance

- Enable partial belonging
Facilitate smooth transitions



- Selected Shapes: Triangular (trimf) & Trapezoidal (trapmf)
- Simplicity, computational efficiency, and intuitive representation for control variables.

FUZZY RULE BASE: GUIDING THE PENDULUM

STRUCTURE:

- IF (INPUT CONDITION 1) AND (INPUT CONDITION 2)
- THEN (OUTPUT ACTION)

ILLUSTRATIVE RULES (EXAMPLES):

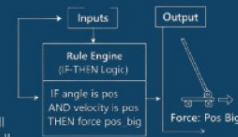
- IF ANGLE IS ZERO AND VELOCITY IS ZERO THEN FORCE IS ZERO
- IF ANGLE IS POS AND VELOCITY IS ZERO THEN FORCE IS POS_SMALL
- IF ANGLE IS POS AND VELOCITY IS POS THEN FORCE IS POS_BIG
- IF ANGLE IS NEG AND VELOCITY IS NEG THEN FORCE IS NEG_BIG

INTUITION:

- TRANSLATES CONTROL HEURISTICS INTO A LOGICAL FRAMEWORK.
- EXAMPLE: IF PENDULUM IS LEANING POSITIVE AND MOVING POSITIVE, APPLY A LARGE POSITIVE FORCE TO PUSH IT BACK.

Fuzzy Rule Base: Guiding the Pendulum

- Structure
 - IF (Input Condition 1) AND THEN (Output Action)
- Illustrative Rules (Examples):
 - IF angle is zero AND velocity is zero THEN force is zero
 - IF angle is pos AND velocity is pos_small THEN force pos_big
 - IF angle is pos AND pos THEN pos_small
 - IF angle is neg neg pos THEN pos_big neg_big



- Intuition: If pendulum leaning Right & falling Right, Right, push Right hard.

QUANTUM COMPUTING: A NEW PARADIGM

CLASSICAL BITS VS. QUANTUM QUBITS:

- **BITS:** HOLD 0 OR 1.
- **QUBITS:** CAN HOLD 0, 1, OR A SUPERPOSITION OF BOTH SIMULTANEOUSLY.
 - THIS ENABLES PARALLEL COMPUTATION.

KEY QUANTUM PHENOMENA:

- **SUPERPOSITION:** QUBITS EXIST IN MULTIPLE STATES AT ONCE.
- **ENTANGLEMENT:** LINKED QUBITS SHARE A FATE, REGARDLESS OF DISTANCE.

QUANTUM GATES:

- OPERATIONS SIMILAR TO CLASSICAL LOGIC GATES (AND, OR, NOT).
- MANIPULATE QUBIT SUPERPOSITIONS AND ENTANGLEMENTS.

RELEVANCE TO FUZZY LOGIC:

- **POTENTIAL FOR PARALLEL INFERENCE:** SUPERPOSITION COULD EVALUATE MULTIPLE FUZZY RULES CONCURRENTLY.
- **NOVEL ARCHITECTURES:** EXPLORES NEW WAYS TO IMPLEMENT COMPLEX AI AND CONTROL LOGIC.



THE QUANTUM FUZZY INFERENCE ENGINE (QFIE)

WHAT IT IS:

- A SPECIALIZED FRAMEWORK (LIBRARY) THAT MAPS CONVENTIONAL FUZZY INFERENCE SYSTEMS ONTO QUANTUM CIRCUITS.
- ENABLES FUZZY LOGIC OPERATIONS TO BE EXECUTED ON QUANTUM HARDWARE OR SIMULATORS.

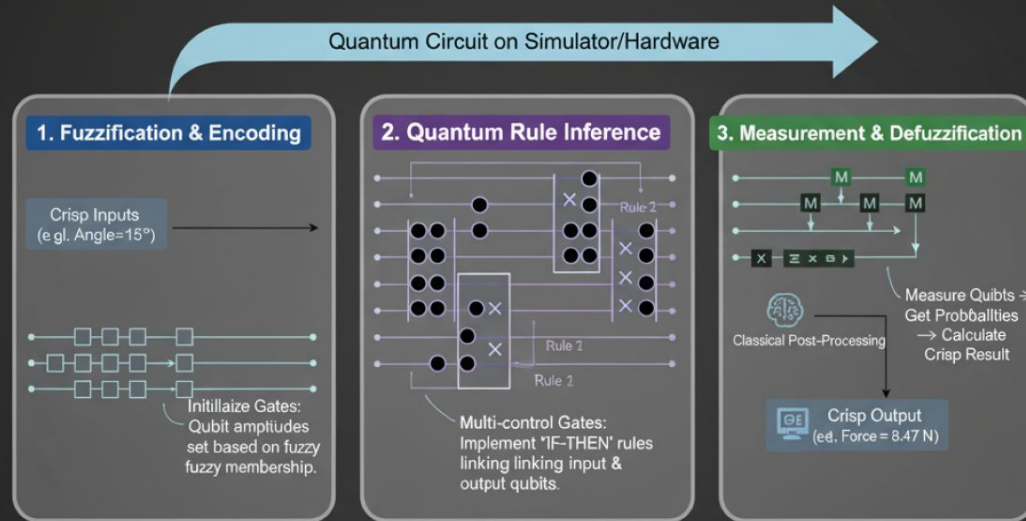
HOW QFIE WORKS (QUANTUM IMPLEMENTATION):

1. FUZZIFICATION: CRISP INPUTS ARE ENCODED INTO QUBIT INITIALIZATION STATES (AMPLITUDES/PROBABILITIES).
2. RULE INFERENCE: FUZZY IF-THEN RULES ARE CONSTRUCTED USING MULTI-CONTROL QUANTUM GATES (E.G., CNOT, TOFFOLI) TO LINK INPUT AND OUTPUT QUBITS.
3. DEFUZZIFICATION: QUANTUM CIRCUIT MEASUREMENTS YIELD PROBABILITIES, WHICH ARE THEN PROCESSED TO DERIVE THE FINAL CRISP OUTPUT.

KEY ADVANTAGE:

- UNLOCKS THE POTENTIAL FOR PARALLEL RULE EVALUATION ON QUANTUM SYSTEMS.

QFIE: Quantum Implementation Workflow



QFIE IMPLEMENTATION: SETUP AND CRISP INPUTS

INITIALIZATION & CONFIGURATION:

Python

```
qfie = QuantumFuzzyEngine(verbose=True, encoding='linear')
qfie.input_variable(name='pendulum_angle', ...)
qfie.output_variable(name='cart_force', ...)
qfie.add_input_fuzzysets(var_name='pendulum_angle', ...)
qfie.add_output_fuzzysets(var_name='cart_force', ...)
qfie.set_rules(rules)
```

EXPLANATION: DEFINES VARIABLES, ASSOCIATES FUZZY SETS, AND LOADS THE RULE BASE.

CRISP INPUT FOR DEMONSTRATION:

Python

```
crisp_inputs = {'pendulum_angle': 15, 'angular_velocity': 2}
```

SCENARIO: PENDULUM LEANING 15° RIGHT, FALLING 2°/S RIGHT.

QFIE Implementation: Setup and Crisp Inputs

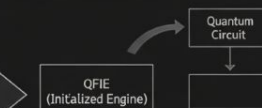
Initialization & Configuration

```
qfie = QuantumFuzzyEngine(verbose=True, encoding='linear')
qfie = input_variable(name='pendulum_angle', ...)

qfie = input_variable(name='pendulum'cart_force)...
{
  {
    out-input-cartforce(...)
    add-out-input fuzzysets(... ..)
    add-ouput fuzzysets(.....)
    set.rules(rule)
  }
}
```

Explanation: Defines variables, associates fuzzy sets, and loads the rule base

Crisp Input for Demonstration



Scenario: Pendulum leaning 15° right, falling 2°/s right

CODE:

- `QFIE.BUILD_INFERENCE_QC(CRISP_INPUTS, DRAW_QC=TRUE, DISTRIBUTED=FALSE)`
- GENERATES THE QUANTUM CIRCUIT BASED ON THE QFIE SETUP AND SPECIFIC CRISP INPUTS.

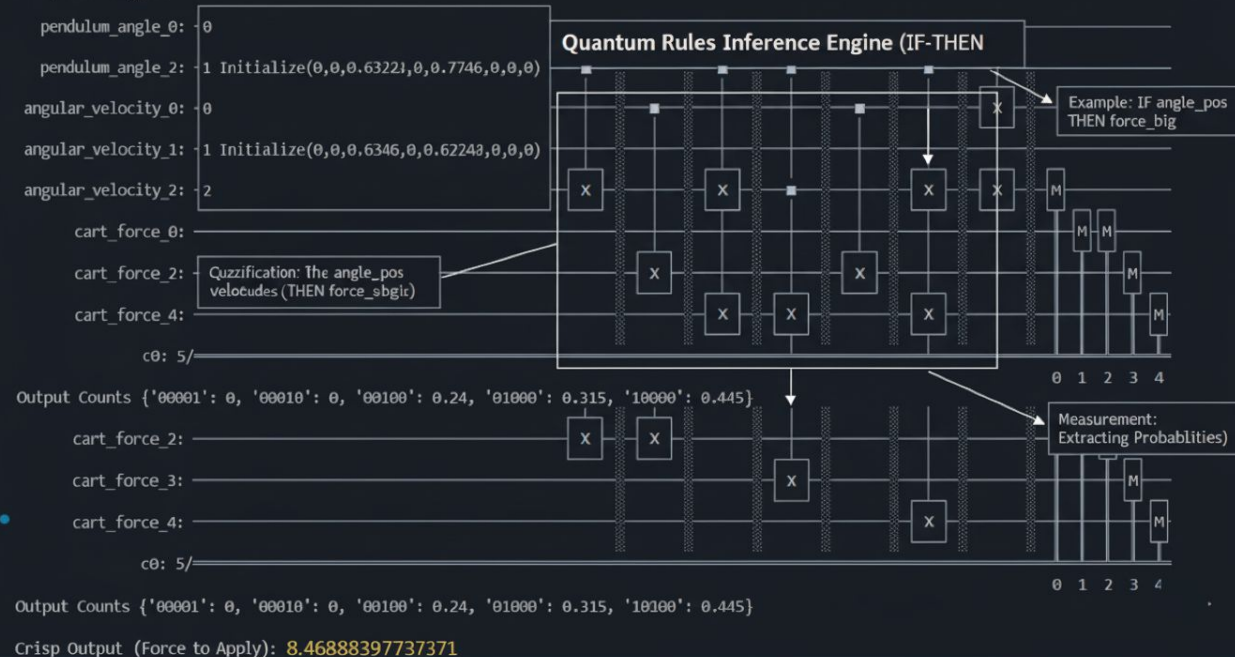
VISUAL REPRESENTATION OF THE QFIE:

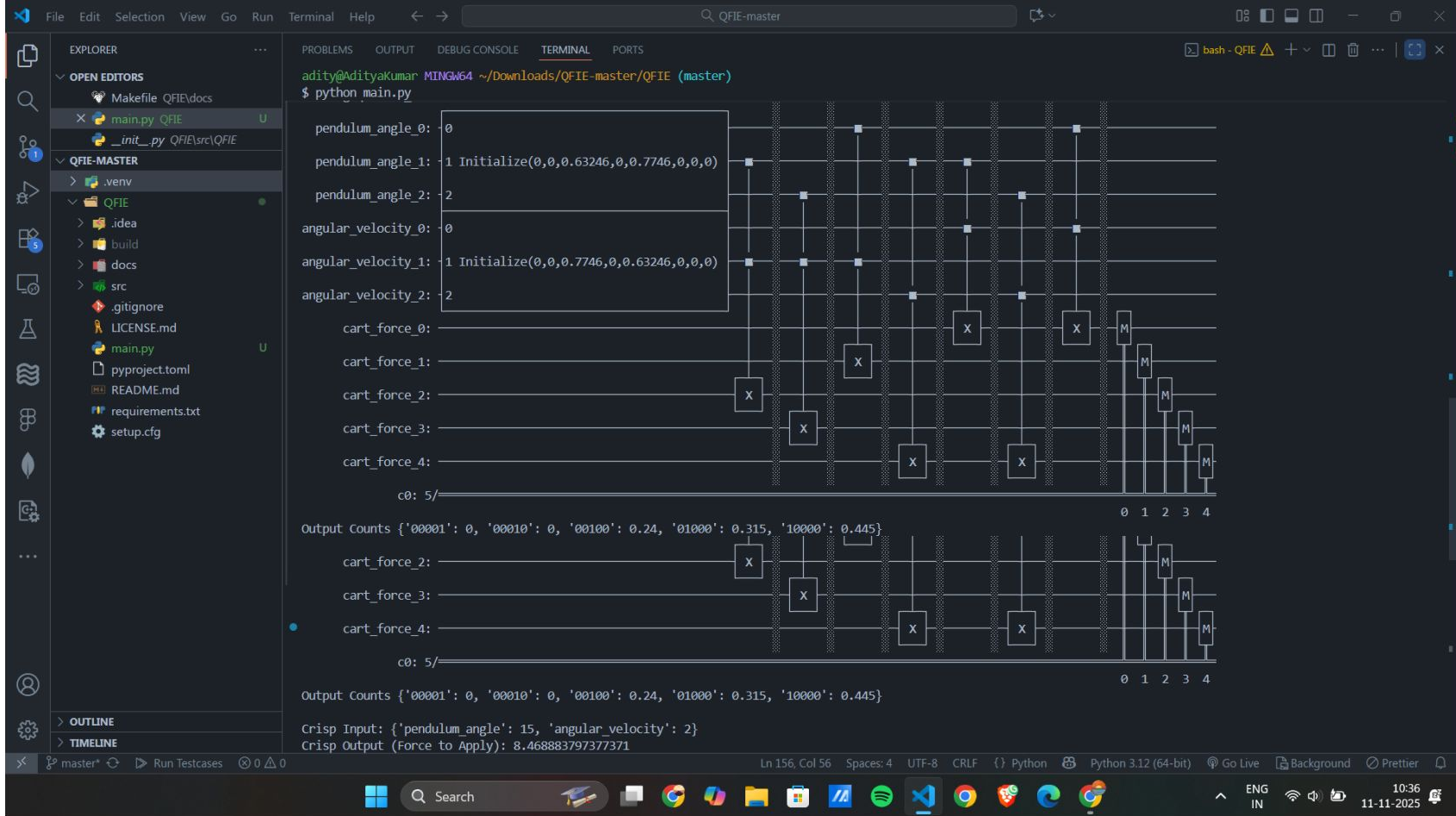
- THIS CIRCUIT DIRECTLY IMPLEMENTS THE FUZZY INFERENCE PROCESS USING QUANTUM GATES.

Quantum Circuit Aesnulator

Code: `qfie.build_inference(qc(crisp=True, distributed=False))`

- Generates the quantum circuit and QFIE setup and specific crisp inputs.
- Visual Representation of the QFIE setup: This circuit directly implements the fuzzy inference process using quantum gates.





FUZZIFICATION OUTPUT & SIMULATION EXECUTION

SIMULATION TRIGGER:

Python

```
crisp_output, activation_values = qfie.execute(n_shots=1000, plot_histo=True)
```

INITIATES QUANTUM CIRCUIT EXECUTION ON THE AERSIMULATOR (1000 SHOTS).

FUZZIFICATION OUTPUT (INPUT VALUES):

```
{'pendulum_angle': [np.float64(0.0), np.float64(0.4), np.float64(0.6)],  
 'angular_velocity': [np.float64(0.0), np.float64(0.6), np.float64(0.399...)]}
```

INTERPRETATION: OUR CRISP INPUTS (ANGLE=15°, VELOCITY=2°/S) ARE CONVERTED.

- ANGLE=15°: 40% ZERO, 60% POS.
- VELOCITY=2°/S: 60% ZERO, 40% POS.

Fuzzification Output & Simulation Execution

Simulation Trigger

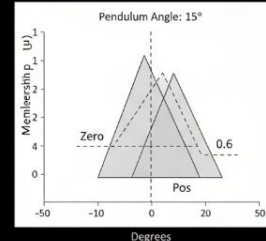
```
crisp_output, activation_values =  
qfie.execute(n_shots=1000,  
plot_histo=True)  
)
```

Initiates quantum circuit execution on the AerSimulator (1000 shots).

Fuzzification Output (Input Values):

```
"[pendulum_angle]:  
  np.float64(0.0) [np.float64(0.4), np.float64(0.6)]  
angular_velocity:  
  np.float64(0.0) [np.float64(0.6, 6.6°, 0.399...)]
```

Interpretation: Our crisp inputs (angle=15°) velocity2), are converted.
angle=15°: 40% zero, 60% pos.
velocity2/s: 60% zero, 40% pos.



QUANTUM INFERENCE RESULTS: OUTPUT COUNTS

RAW SIMULATION OUTCOMES:

OUTPUT COUNTS {'00001': 0, '00010': 0, '00100': 0.24, '01000': 0.315, '10000': 0.445}

INTERPRETATION:

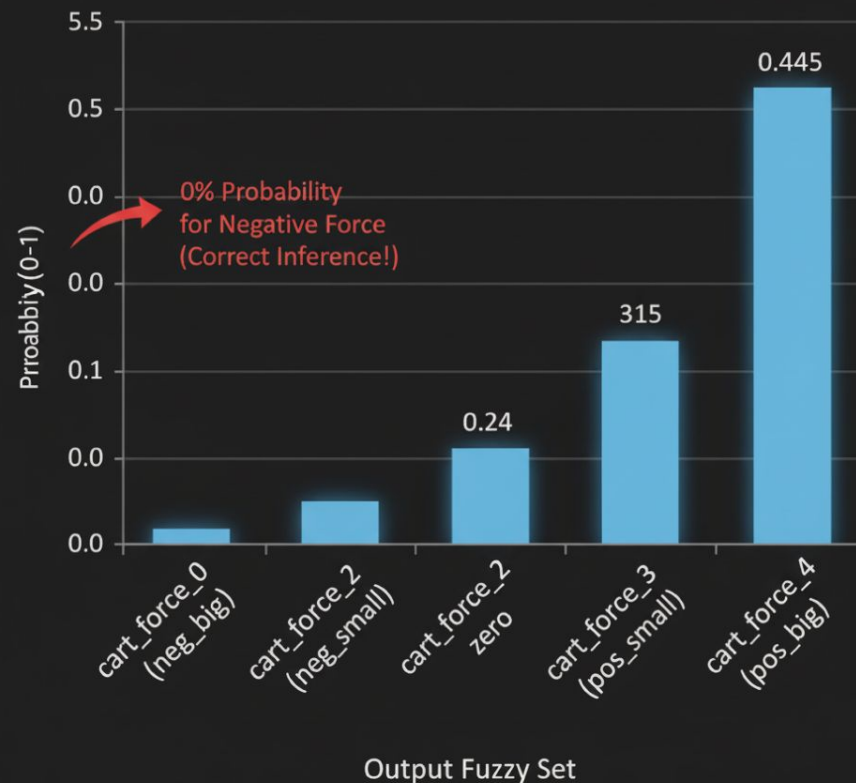
- THESE ARE THE MEASURED PROBABILITIES (FREQUENCIES) OF ACTIVATED OUTPUT FUZZY SETS FROM 1000 QUANTUM CIRCUIT EXECUTIONS.
- EACH BINARY STRING ('10000', '01000', ETC.) CORRESPONDS TO A SPECIFIC OUTPUT FUZZY SET.

KEY INSIGHT:

- '10000' (FORCE POS_BIG): 44.5%
- '01000' (FORCE POS_SMALL): 31.5%
- '00100' (FORCE ZERO): 24%
- CRUCIALLY, NEGATIVE FORCE OUTPUTS ('00001', '00010') REGISTERED 0%, CONFIRMING APPROPRIATE INFERENCE FOR THE POSITIVE INPUT.

Inference Output Counts & Probabilities

Output Counts ("00001": 0, "00100 0.24, 315, 0.445)



DEFUZZIFICATION: THE FINAL CONTROL ACTION

PROCESS:

- COMBINES THE MEASURED FUZZY OUTPUT PROBABILITIES (FROM OUTPUT COUNTS).
- CONVERTS THESE INTO A SINGLE, UNAMBIGUOUS CRISP OUTPUT VALUE (THE CONTROL COMMAND).
- TYPICALLY USES METHODS LIKE THE "CENTER OF GRAVITY" (COG) FOR PRECISE ACTION.

CALCULATED CRISP OUTPUT:

Crisp Output (Force to Apply): 8.468883797377371

INTERPRETATION:

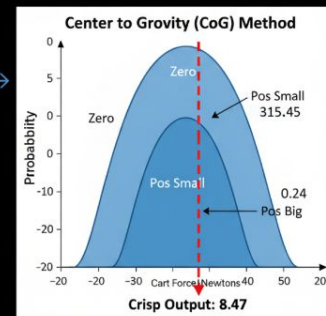
- BASED ON $\text{PENDULUM_ANGLE}=15^\circ$ AND $\text{ANGULAR_VELOCITY}=2^\circ/\text{s}$,
- THE QFIE DETERMINES AN EXACT FORCE OF ~ 8.47 NEWTONS MUST BE APPLIED TO THE CART.
- THIS IS THE PRECISE, ACTIONABLE COMMAND TO STABILIZE THE INVERTED PENDULUM.

Defuzzification: The Final Control Action

Calculated Crisp Output

Crisp Output (Force to Apply):
8.46888379 733777371

Interpretation: Combines fuzzy probabilities into a single, actionable value
QFIE determines an exact force of ~ 8.47 Newtons must be applied to stabilize the pendulum.

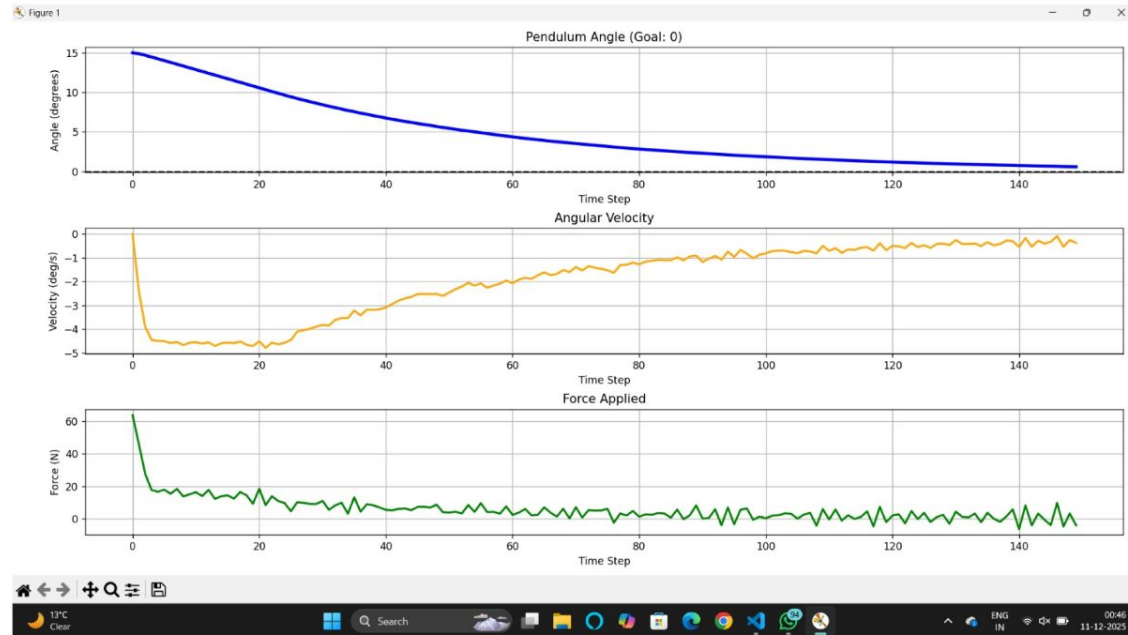


The CoG method calculates the weighted average force

OUTPUT AFTER 150 ITERATION

OBSERVATIONS:

- THE PENDULUM BEGINS AT $+15^\circ$ AND DECAYS SMOOTHLY TOWARD 0° , INDICATING CORRECT STABILIZING ACTION.
- ANGULAR VELOCITY BECOMES NEGATIVE INITIALLY, SHOWING THAT THE CONTROLLER APPLIES TORQUE TO COUNTERACT THE POSITIVE TILT.
- THE APPLIED FORCE PEAKS HIGH AT THE FIRST STEP AND THEN GRADUALLY DECREASES AS THE PENDULUM APPROACHES EQUILIBRIUM.
- THE MOTION IS WELL-DAMPED WITH NO OSCILLATORY DIVERGENCE.
- THE FUZZY RULES "ANGLE POSITIVE $\Rightarrow$ APPLY POSITIVE FORCE" PERFORM AS INTENDED.

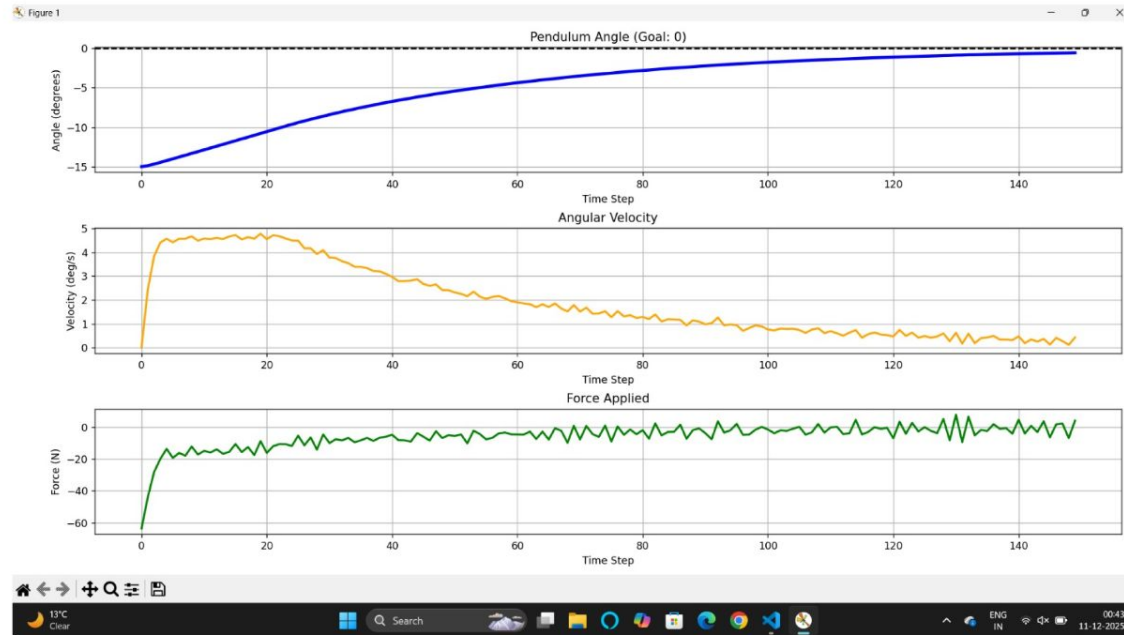


Output Plot of Case 1 :- Initial Angle $+15^\circ$

OUTPUT AFTER 150 ITERATION

OBSERVATIONS:

- THE PENDULUM BEGINS AT -15° AND MOVES UPWARD TOWARD 0° IN A STABLE MANNER.
- ANGULAR VELOCITY BECOMES STRONGLY POSITIVE EARLY ON, WHICH IS CONSISTENT WITH THE REQUIRED TORQUE TO CORRECT A NEGATIVE TILT.
- THE APPLIED FORCE SHOWS A LARGE NEGATIVE VALUE INITIALLY, THEN SETTLES TOWARD SMALLER CORRECTIVE VALUES.
- THE CONTROLLER HANDLES THE SYMMETRIC CASE WITH STABILITY AND WITHOUT OVERSHOOT.
- THE “ANGLE NEGATIVE $\Rightarrow$ APPLY NEGATIVE FORCE” RULES PRODUCE SMOOTH CORRECTIVE ACTION



Output Plot of Case 2 :- Initial Angle -15°

CONCLUSION & FUTURE SCOPE

CONCLUSION:

- SUCCESSFULLY DEMONSTRATED A QUANTUM FUZZY INFERENCE ENGINE (QFIE) FOR STABILIZING AN INVERTED PENDULUM.
- VALIDATED THE FUSION OF FUZZY LOGIC AND QUANTUM COMPUTING FOR DYNAMIC SYSTEM CONTROL.

KEY ACHIEVEMENT:

- PIONEERING THE IMPLEMENTATION OF QUANTUM-ENHANCED FUZZY INFERENCE FOR A NON-LINEAR CONTROL CHALLENGE.
- SHOWCASES THE VIABILITY OF NOVEL COMPUTATIONAL PARADIGMS IN INTELLIGENT CONTROL SYSTEMS.

POTENTIAL BENEFITS:

- ENHANCED SCALABILITY FOR HIGHLY COMPLEX RULE-BASED SYSTEMS.
- OPENS AVENUES FOR EXPLORING TRULY QUANTUM-NATIVE CONTROL ALGORITHMS.

FUTURE WORK:

- HARDWARE IMPLEMENTATION: DEPLOYING QFIE ON ACTUAL QUANTUM PROCESSORS.
- OPTIMIZATION: FURTHER RESEARCH INTO QUANTUM CIRCUIT COMPLEXITY AND EFFICIENCY.
- ADVANCED ARCHITECTURES: INVESTIGATING NOVEL ENCODING AND RULE-MAPPING SCHEMES.
- EXPANDED APPLICATIONS: APPLYING QFIE TO MORE INTRICATE CONTROL PROBLEMS (E.G., MULTI-JOINT ROBOTICS, ADAPTIVE SYSTEMS).

THANK YOU

THANK YOU